

Network Intrusion Detection Using Zeek

Problem Statement

Network intrusion is still an immense challenge for people and companies; it is indiscriminate. Attackers continue to increase the pressure as they improve their attacks. Many companies and individuals still lack the ability to detect threats. This project is going to close that gap between the attackers and all of the other people at risk. This project will use Zeek to monitor and log patterns in live network traffic. This will be a reproducible pipeline that demonstrates meaningful network coverage without needing an expensive solution.

Threat Model

The assets are a target server virtual machine that represents a normal small network host. Zeek and the logs are the other smaller assets. The potential attackers are simulated attackers coming from a Kali Linux virtual machine on the same network. The attack scenarios include but are not limited to network recon, DDOS floods, and basic exploitation. The attack surfaces are on the exposed network services of the virtual machine. The assumptions are that all the traffic is unencrypted or using standard TLS. The defensive approach is that Zeek will monitor all the traffic on a port. Custom Zeek scripts will implement detection rules for the attacks and produce alerts when certain thresholds are met.

Success Metrics

This is an example of the metrics, but false positive rates should be less than 5% of the total alerts. As well as the detection rate being at least 90% accurate of true positive attacks.

Dataset

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All data will be captured within a virtual machine home lab. There will be no external traffic used. Traffic captured will be put into labeled sections with the specific intrusion. All of the data is generated on the virtual machine, and no real user data is involved.

Risks and Ethics

All attacks are in an isolated virtual machine with no external connection. There will be no real users or data are affected. Since all of the data is self-generated, there's no risk of real data being captured. The risk of this project is that the data may not really replicate actual network conditions. This can potentially lead to limits in the detection of results.

Architecture Diagram

